

DAY 10 - LINUX FILES & PERMISSIONS

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1. What is a Linux directory?

A Linux directory is a folder used to organize files and other directories in a hierarchical filesystem. Directories help users and programs locate and manage data.

2. What does mkdir do?

mkdir creates a new directory. For example, ``mkdir day10_lab`` creates a directory named day10_lab in the current location.

3. What does touch do?

touch creates an empty file if the specified file does not exist. If the file already exists, it normally updates its timestamps without changing its contents.

4. What does cat do?

cat displays the contents of a file in the terminal. It can also be used to combine files or redirect content, depending on how it is used.

5. What does cp do?

cp copies files or directories from one location to another. The original source remains in place unless it is separately removed.

6. What does mv do?

mv moves a file or directory to another location. It is also commonly used to rename files or directories.

7. What does rm do?

rm removes files. It should be used carefully because deleted files may not be recoverable through normal filesystem operations. The lab instruction specifically says to remove only files created for the exercise.

8. What does ls -l display?

ls -l displays a detailed listing of files and directories, including permissions, link count, owner, group, size, modification time, and name.

9. Explain r, w and x permissions.

r means read: a file can be read. w means write: a file can be modified. x means execute: a file can be executed as a program or script. For directories, these permissions have directory-specific meanings.

10. Explain user, group and others.

User (u) refers to the owner of the file. Group (g) refers to users belonging to the file's assigned group. Others (o) refers to all other users who are neither the owner nor members of that group.

11. What does chmod do?

chmod changes the permission bits of a file or directory. It can use symbolic notation such as ``u+x`` or numeric/octal notation such as ``644`` and ``755``.

12. What does chmod 755 represent?

755 means: owner has read, write and execute (7); group has read and execute (5); others have read and execute (5). Symbolically, this is commonly ``rwxr-xr-x``.

13. What does chmod 644 represent?

644 means: owner has read and write (6); group has read only (4); others have read only (4). Symbolically, this is commonly ``rw-r--r--``.

14. What is file ownership?

File ownership identifies the user who owns a file and the group associated with it. Ownership works together with permissions to control access.

15. What is the purpose of /etc?

The /etc directory contains system-wide configuration files and configuration information used by the operating system and installed services.

16. What is the purpose of /home?

The /home directory normally contains personal directories and files for regular users, such as documents, downloads, and user-specific configuration data.

17. What is commonly stored under /var?

The /var directory contains variable data that changes during system operation, such as logs, caches, spools, databases, and other application/runtime data.

18. Why should rm be used carefully?

rm can permanently remove files and normally does not provide a recycle-bin step. Using it with the wrong path, wildcard, or privileges can delete important data, so it should be restricted to files created for the lab.

19. Why are Linux permissions important?

Permissions provide access control by limiting who can read, modify, or execute files and directories. They help protect data, system configuration, and applications from unauthorized access or modification.

20. How can incorrect permissions create a security risk?

Overly permissive permissions can allow unauthorized users to read sensitive files, modify important data, or execute programs. For example, a sensitive file that is readable by everyone may expose confidential information.